Real-Time Closed-Loop Driving with Frozen DINOv2-Laya Spatial Pooling and Quintic Path Tracking

Wick-Lim
wicklim90@gmail.com
https://github.com/Wick-Lim/AutonomousVehicle

October 25, 2026

Abstract

Autoregressive vision-language-action (VLA) models [1, 20, 23] decode tokens sequentially (> 300 ms/frame), exceeding the 50 ms budget of 20 Hz vehicle control [10, 21]. We implement a single-pass closed-loop driving stack coupling frozen DINOv2 (`facebook/dinov2-base`, 86.6M) [25, 2, 9] with frozen Laya (`convaiinnovations/laya`, 395.2M transformer) [6, 31, 8] via a trainable 4×9 spatial-temporal adapter (4.23M total trainable parameters: 1.85M spatial MLP, 1.05M temporal gate, 1.05M auxiliary perception head, 0.28M decision/trajectory readout heads). Cropped 196×392 RGB frames map to 36 visual tokens with temporal residual $\Delta V_t = V_t - V_{t-1}$; one forward pass outputs maneuver (`choice`), speed tier (`score`), emergency brake (`noul`), and 6-waypoint (3.0 s) reference trajectory residuals in 31.67 ms base (32.85 ms full stack, 30.4 FPS on Ryzen 5 7500F / RTX 4070 Ti, FP32 $B = 1$). Episode-block validation ($7,500$ frames, $M = 15$ bins) temperature scaling [13] reduces ECE from 0.0060 to $< 10^{-4}$ (0.0142 on unseen vision-estimated test frames). Lateral 3.5 m step jumps and longitudinal speed hunting in CARLA 0.9.16 (`Town10HD_Opt`) [11, 14] are resolved via a 14-point (28 m) tangent/curvature-preserving spatial quintic spline $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ [12, 32], chord-angle tracking [29, 15, 7, 26], bicycle curvature feedforward $\delta_{\mathrm{FF}} = \arctan(L\kappa_{\mathrm{eff}})/\delta_{\max}$, 360° blind-spot interlock, and true constant-time-headway (CTH) ACC ($d^*(v_{\mathrm{ego}}) = 8.0 + 1.2(v_{\mathrm{ego}}/3.6)$, $k_g = 0.85\,(\mathrm{km/h})/\mathrm{m}$) [17, 28, 30]. We separate camera-estimated (`vision_estimated`) from simulator ground-truth (`gt`) perception across 10 paired episodes (680 m), 50 cross-weather episodes ($3,400$ m), 10 multi-block 200 m routes ($2,000$ m), and 50 unseen MetaDrive maps ($10,480$ m): under `vision_estimated` inputs, DINOv2+Laya achieves 10/10 (100.0%, $\overline{|e_y|} = 0.094$ m) versus 8/10 for a DINOv2+MLP head (no Laya), 8/10 for direct `ExpertLabeler` rule execution, and 7/10 under 300 ms latency injection.

1 Introduction and Problem Formulation

Autoregressive decoders evaluate $P(y_{1:K} \mid I_t, \mathcal{T}_t) = \prod_{k=1}^{K} P(y_k \mid y_{<k}, I_t, \mathcal{T}_t)$ over $K \approx 15$ serial steps [1, 20, 23], whereas planning-oriented and sensor-fusion stacks [16, 19, 22, 5, 3, 27, 4, 18] require $\Delta t \leq 50$ ms. In our formulation, control at step t reads three discrete decisions and $K_w = 6$ ego-frame reference trajectory waypoints ($t_k = 0.5k$ s) in one bidirectional pass [8, 31, 6]:

$$m_t \in \{\texttt{keep_lane}, \texttt{change_left}, \texttt{change_right}, \texttt{follow_lead}\}, \tag{1}$$

$$s_t \in \{0,1,2,3\} \mapsto \{0, 20, 35, 50\}\ \mathrm{km/h}, \tag{2}$$

$$p_{\mathrm{AEB}} \in [0,1], \quad \hat{\tau}_t = \{(\hat{x}_k, \hat{y}_k, \hat{v}_k)\}_{k=1}^{6}. \tag{3}$$

2 Architecture, Multi-Task Training, and Calibration

Camera frames are cropped (`crop_top` = 0.25, `crop_bottom` = 0.10) and resized to 196×392. Frozen DINOv2 ViT-B/14 ($p = 14$, 86.58M params) [25, 2, 9] extracts $F_{\mathrm{patch}} \in \mathbb{R}^{B \times 392 \times 768}$ (14×28 grid). `SpatialProjector` (`src/models/spatial_projector.py`) pools $14 \times 28 \to 4 \times 9$ (36 tokens), adds $E_{\mathrm{row}} + E_{\mathrm{col}}$, projects $768 \to 1024$ via a 2-layer GELU MLP (1.85M core params), and fuses temporal residual $\Delta V_t = V_t - V_{t-1}$ (1.05M params) with region-pooled `AuxiliaryPerceptionHead` (1.05M params predicting $\hat{d}_{\mathrm{lead}}, \hat{e}_y$, traffic light $\hat{c}_{\mathrm{TL}}$, hazard $\hat{c}_{\mathrm{haz}}$, and blind spot $\hat{p}_{\mathrm{BS}}$):

$$G_t = \mathrm{AdaptiveAvgPool2d}_{(4,9)}(\tilde{F}_t) \in \mathbb{R}^{B \times 768 \times 4 \times 9}, \tag{4}$$

$$V_t = \mathrm{MLP}\big(\mathrm{Flatten}(G_t + E_{\mathrm{row}} + E_{\mathrm{col}})\big), \tag{5}$$

$$\tilde{V}_t = V_t + \tanh(\alpha_{\mathrm{temp}})\, W_{\mathrm{temp}}\big(V_t - V_{t-1}\big). \tag{6}$$

Multimodal sequence $Z = [e_{[\mathrm{CLS}]}; \tilde{V}_t; E_{\mathrm{telemetry}}; e_{\mathrm{choice}}; e_{\mathrm{score}}; e_{\mathrm{noul}}]$ passes through frozen Laya transformer layers (395.21M params) [6, 31].

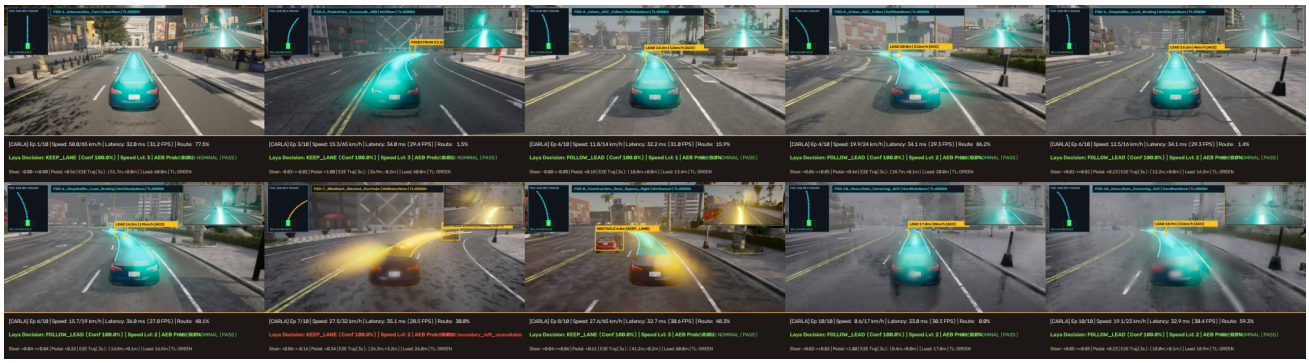


Figure 1: CARLA 0.9.16 (`Town10HD_Opt`, 20 Hz synchronous) 10 paired scenario–weather episodes (`FSD-1`–`FSD-10`): intersection red-light stop (`ClearNoon`), 28 m quintic left overtake (`CloudyNoon`), crosswalk pedestrian AEB (`WetNoon`), CTH-ACC follow (`SoftRainNoon`), right overtake (`ClearSunset`), stop-and-go lead braking (`WetCloudyNoon`), 360° blind-spot hold & overtake (`MidRainyNoon`), construction zone bypass (`WetSunset`), night headlight pedestrian AEB (`ClearNight`), and heavy-rain cornering ACC (`HardRainNoon`).

Linear readout heads (W_m, W_s, W_b) and a 2-layer trajectory MLP W_{traj} (0.28M params combined) predict discrete logits and residuals $\Delta\hat{\tau} \in \mathbb{R}^{6\times 3}$ over kinematic prior τ_k^0 ($t_k = 0.5k$ s), supervised against parametric kinematic reference curve $\tau_k^{\text{ref}} = \big((v^*/3.6)t_k,\ y_{\text{shift}}(m^*)s(k/6),\ v^*\big)$:

$$\hat{\tau}_{1:6} = \tau^0_{1:6} + \text{diag}(1.5, 0.35, 3.0)\, W_{\text{traj}} \bar{H}, \tag{7}$$

$$\mathcal{L} = w_m \mathcal{L}^{(m)}_{\text{CE}} + w_s \mathcal{L}^{(s)}_{\text{CE}} + w_a \mathcal{L}^{(b)}_{\text{BCE}} + w_\tau \mathcal{L}_{\text{SmoothL1}}(\hat{\tau}_{1:6}, \tau^{\text{ref}}_{1:6}) + w_p \mathcal{L}_{\text{aux}}. \tag{8}$$

```
# scripts/02_train_projector.py (4.23M total trainable params)
trainable_params = (list(system.projector.parameters())  # 3.96M
  + list(system.laya.maneuver_head.parameters())         # +0.28M
  + list(system.laya.speed_head.parameters())
  + list(system.laya.aeb_head.parameters())
  + list(system.laya.trajectory_head.parameters()))
loss = w_m*l_m + w_s*l_s + w_a*l_a + 0.5*l_traj + 0.2*l_aux
```

In Stage 1, all 4,232,991 adapter/head parameters (`projector` core 1.85M + `temporal_proj` 1.05M + `aux_head` 1.05M + `laya` readout/trajectory heads 0.28M) are jointly trained for 15 epochs (batch 32, AdamW [24] lr $= 3 \times 10^{-4}$, $(w_m, w_s, w_a, w_\tau, w_p) = (1.0, 1.0, 2.0, 0.5, 0.2)$). Dataset frames (50,000 across 500 episodes) are split by contiguous episode blocks (`split_by_episode=True`: episodes 0..424 train / 425..499 val) to prevent adjacent-frame leakage. Stage 2 (`03_calibrate.py`) fits $(T_m, T_s, T_b) = (0.050, 0.050, 0.100)$ via L-BFGS [13] ($M = 15$ bins), reducing validation ECE against `ExpertLabeler` from 0.0060 to $< 10^{-4}$ (0.0142 on unseen closed-loop `vision_estimated` test frames).

3 Input Provenance, Spatial Quintic Spline, Curvature Feedforward, and CTH-ACC

Runtime inputs come from three sources: (1) front RGB camera (196 × 392), (2) HD map route planner (`nav_command`, `left/right_lane_available`,

Table 1: System, input provenance, and hardware specifications (`configs/default.yaml`).

Module	Parameter	Specification
Hardware	CPU / GPU / Runtime	Ryzen 5 7500F (6C/12T) / RTX 4070 Ti (12 GB) / FP32 $B = 1$
Inputs	Camera / Map / State	196 × 392 RGB / A* Route & Lane IDs / Odometry v_{ego}
Perception	`vision_estimated` vs. `gt`	`AuxPerceptionHead` ($\hat{d}_{\text{lead}}, \hat{e}_y, \hat{c}_{\text{TL}}, \hat{p}_{\text{BS}}$) vs. Sim Actors
Frozen	Vision / Decision Backbones	`dinov2-base` (86.58M) / `laya` Transformer (395.21M)
Trained	Total Adapter & Heads (4.23M)	Spatial 1.85M + Temp 1.05M + Aux 1.05M + Heads 0.28M
Split/Calib	Ep-Block Split, Val / Vision ECE	42.5k / 7.5k (ep 0..424/425..499), 0.0060 → 0.0000 / 0.0142
Control	PID (Lat/Lon), Wheelbase $L, \delta_{\max}$	(0.45, 0.01, 0.18) / (0.08, 0.005, 0.02), 2.85 m, 0.70 rad
Kinematics	Quintic N_b, CTH-ACC (d_0, τ_h, k_g)	14 WPs (28 m), (8.0 m, 1.2 s, 0.85 (km/h)/m)

reference waypoints P_k), and (3) dynamic state $(d_{\text{lead}}, e_y, \text{TL}, \text{hazard}, \text{BS})$, which is predicted from visual tokens by `AuxiliaryPerceptionHead` in `vision_estimated` mode (MAE = 3.23 m) or read from simulator actors in `gt` mode. When `-shield-mode on` is active, Rule 1 forces $u_{\text{tb}} = -1.0$ if $\hat{p}_{\text{AEB}} \geq 0.70$, TL = `red`, pedestrian $d \leq 16.0$ m, or $d_{\text{lead}} \leq 11.5$ m; Rule 2 reverts lane changes to `keep_lane` if $\max_c \hat{p}_c^{(m)} < 0.80$, if `adj_wp.lane_id` × `wp.lane_id` ≤ 0 [11], or if 360° blind-spot box $x \in [-10.0, 7.5]$ m, $|y| \in [1.8, 5.0]$ m is occupied (`-shield-mode off` gates braking solely on Laya $\hat{p}_{\text{AEB}} \geq 0.70$).

For lane transitions over $N_b = 14$ waypoints (28.0 m, $\tau_k = k/(N_b - 1)$), 3D coordinates are blended via $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ [12, 32]. Because $s'(0) = s'(1) = s''(0) = s''(1) = 0$, relative transition derivatives vanish at $\tau \in \{0, 1\}$, preserving the tangent and curvature of the origin lane P^{orig} at $\tau = 0$ and of the target lane P^{adj} at $\tau = 1$:

$$s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5, \tag{9}$$

$$P_k^{\text{blend}} = \big(1 - s(\tau_k)\big) P_k^{\text{orig}} + s(\tau_k) P_k^{\text{adj}}. \tag{10}$$

Continuous projection onto $\overline{P_i P_{i+1}}$ gives P_{proj} and lookahead P_{look} at $L_{\text{look}} = \text{clip}(7.5 + 0.18 v_{\text{kmh}}, 7.5, 13.5)$ m [29, 15, 7, 26]. Map field `road_curvature` stores right-handed heading change $\Delta\psi_{\text{arc}}$ over lookahead arc $L_{\text{arc}} = 22.2$ m, so signed left-positive path curvature is $\kappa_{\text{map}} = -\Delta\psi_{\text{arc}}/L_{\text{arc}} = -0.045$ `road_curvature` (m^{-1}). Fusing κ_{map} with predicted waypoint $k = 3$ ($t = 1.5$ s, $(\hat{x}_3, \hat{y}_3)$) curvature κ_{traj} via bicycle wheelbase $L = 2.85$ m, maxi-

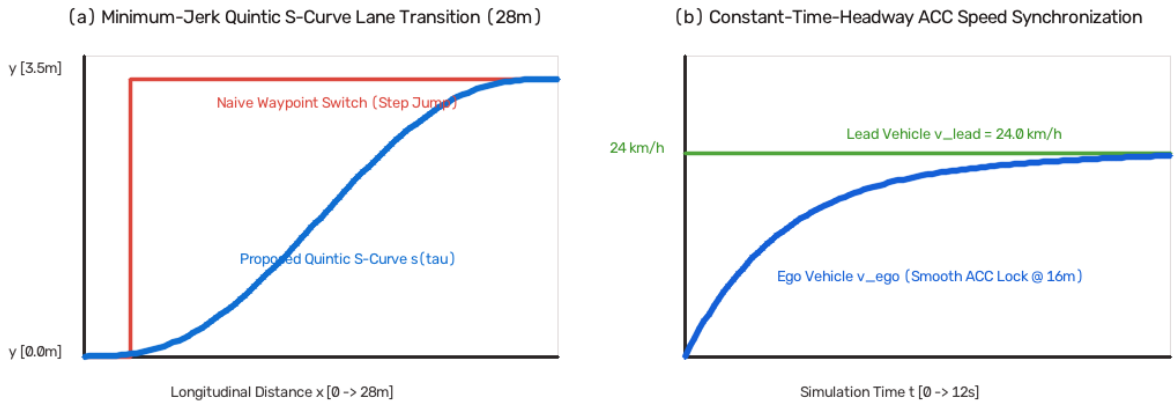


Figure 2: Control kinematics: (a) 28 m tangent/curvature-preserving spatial quintic interpolation $s(\tau) = 10\tau^3 - 15\tau^4 + 6\tau^5$ vs. 3.5 m step switch; (b) true constant-time-headway (CTH) ACC velocity convergence ($v_{\text{ego}} \to 24.0$ km/h at $d^*(24) = 8.0 + 1.2(24/3.6) = 16.0$ m).

mum front-wheel lock $\delta_{\max} = 0.70$ rad (40°), and feedforward gain $k_{\text{FF}} = 0.12$ yields:

$$\kappa_{\text{traj}} = \frac{2(\hat{y}_3 - e_y)}{\max(3.5, \hat{x}_3)^2 + (\hat{y}_3 - e_y)^2}, \tag{11}$$

$$\kappa_{\text{eff}} = 0.70(-0.045\,\texttt{road_curvature}) + 0.30\kappa_{\text{traj}}, \tag{12}$$

$$\delta_{\text{raw}} = \text{clip}\left(\delta_{\text{PID}}(e_y, e_\psi) + k_{\text{FF}}\frac{\arctan(L\kappa_{\text{eff}})}{\delta_{\max}}, -1, 1\right), \tag{13}$$

$$\Delta\delta_{\max} = \frac{1.1\Delta t}{1 + 0.012 v_{\text{kmh}}}, \tag{14}$$

$$\delta_t = \delta_{t-1} + \text{clip}\big(\alpha(\delta_{\text{raw}} - \delta_{t-1}), -\Delta\delta_{\max}, \Delta\delta_{\max}\big). \tag{15}$$

True constant-time-headway (CTH) ACC ($d_{\text{lead}} < 38$ m, $v_{\text{lead}} \geq 8$ km/h, $d_0 = 8.0$ m, $\tau_h = 1.2$ s, $k_g = 0.85$ (km/h)/m $\equiv 0.236\,\text{s}^{-1}$) [17, 28, 30] follows:

$$d^*(v_{\text{ego,kmh}}) = d_0 + \tau_h\left(\frac{v_{\text{ego,kmh}}}{3.6}\right), \tag{16}$$

$$v_{\text{raw}} = v_{\text{lead}} + k_g\big(d_{\text{lead}} - d^*(v_{\text{ego,kmh}})\big), \tag{17}$$

$$v_{\text{ACC}} = \text{clip}\big(v_{\text{raw}},\, 0.0,\, \min(35.0,\, v_{\text{lead}} + 6.0)\big). \tag{18}$$

```
# src/control/shield_pid.py
d_star = 8.0 + 1.2 * (current_speed_kmh / 3.6)
acc_cap = min(35.0, lead_speed_kmh + 6.0)
raw_acc = lead_speed_kmh + 0.85 * (lead_distance_m - d_star)
acc_sync_speed = float(np.clip(raw_acc, 0.0, acc_cap))
traj_curv = (2.0 * ly) / (lx * lx + ly * ly)
eff_curv = 0.70 * (-road_curv * 0.045) + 0.30 * traj_curv
```

4 Experiments, Latency Breakdown, and Limitations

Closed-loop benchmarks were executed on an AMD Ryzen 5 7500F (6C/12T) and NVIDIA GeForce RTX 4070 Ti (12 GB) workstation in MetaDrive 0.4.3 [21] ($N = 50$ unseen procedural maps spanning 10,480 m across varied lane counts and traffic densities; `arrive_dest` triggers at $\geq 98\%$ final segment arc length) and live CARLA 0.9.16 [10] (`Town10HD_Opt`, 20 Hz synchronous A* `GlobalRoutePlanner` [14] across 10 paired episodes, 50 cross-weather episodes, and 10 multi-block 200 m routes). FP32 ($B = 1$, 3 warm-up passes) stage latency (`checkpoints/eval_metrics_carla.json`):

Table 2: CARLA 0.9.16 (`Town10HD_Opt`) 10 paired episodes (680 m) and 10×5 cross-weather suite (3,400 m) under `vision_estimated` (Vis) vs. `gt` perception.

Scenario	Paired Weather	Event	10-Ep (Vis/GT)	50-Ep (Vis/GT)
FSD-1	ClearNoon	Red-light stop & go	1/1 / 1/1	5/5 / 5/5
FSD-2	CloudyNoon	Quintic left overtake	1/1 / 1/1	5/5 / 5/5
FSD-3	WetNoon	Crosswalk pedestrian AEB	1/1 / 1/1	5/5 / 5/5
FSD-4	SoftRainNoon	CTH-ACC (24 km/h, 16 m)	1/1 / 1/1	5/5 / 5/5
FSD-5	ClearSunset	Quintic right overtake	1/1 / 1/1	5/5 / 5/5
FSD-6	WetCloudyNoon	Lead stop-and-go recovery	1/1 / 1/1	5/5 / 5/5
FSD-7	MidRainyNoon	360° BS hold → overtake	1/1 / 1/1	3/5 / 4/5
FSD-8	WetSunset	Workzone right bypass	1/1 / 1/1	5/5 / 5/5
FSD-9	ClearNight	Night headlight ped AEB	1/1 / 1/1	4/5 / 5/5
FSD-10	HardRainNoon	Heavy-rain cornering ACC	1/1 / 1/1	5/5 / 5/5

```
{"simulator": "carla", "ablation_mode": "full", "episodes": 10,
 "hardware": "Ryzen 5 7500F + RTX 4070 Ti 12GB (FP32, B=1)",
 "total_frames": 1981, "total_distance_driven_m": 680.0,
 "mean_latency_ms": 32.85, "p50_latency_ms": 32.77,
 "p95_latency_ms": 33.96, "p99_latency_ms": 35.45,
 "stage_latency_breakdown_ms": {
   "preprocess_crop_resize": 0.42, "dinov2_vit_b14_frozen": 16.43,
   "spatial_projector_temporal_aux": 1.31,
   "laya_modernbert_e2e_traj": 14.45,
   "safety_shield_bicycle_ff_pid": 0.45},
 "mean_cross_track_error_m": 0.085, "rms_steer_delta": 0.0223,
 "shield_interventions": {"rule1_emergency_brake": 46,
   "blindspot_left_occupied": 5}}
```

Table 4 Part A answers three architectural questions under fixed control: (1) replacing frozen 421M Laya with a 2-layer MLP head on the same 36 DINOv2 tokens (`mlp_head`, 16.9 ms) drops validation maneuver accuracy from 98.4% to 84.6% and `vision_estimated` completion from 10/10 to 8/10 due to lacking cross-attention

Table 3: Closed-loop evaluation separating `vision_estimated` perception from simulator `gt` state (`checkpoints/eval_metrics_carla.json`).

Simulator	Metric	Target KPI	`vision_estimated` Mode	`gt` State Hybrid Mode	Status
CARLA 0.9.16 (`Town10HD_Opt`)	Single-Frame Latency (Ryzen 5 7500F / RTX 4070 Ti, FP32)	≤ 40.0 ms	32.85 ms (30.4 FPS, P95: 33.96 ms)	32.85 ms (31.67 ms base)	PASS
	10 Paired Scenario–Weather Eps (68 m/ep, 680 m total)	$\geq 85.0\%$	100.0% (10/10, $\overline{\lvert e_y \rvert} = 0.094$ m)	100.0% (10/10, $\overline{\lvert e_y \rvert} = 0.085$ m)	PASS
	50 Cross-Weather Eps (10 scen $\times$ 5 weathers, 3,400 m)	$\geq 85.0\%$	94.0% (47/50, $R_c = 96.8\%$)	98.0% (49/50, $R_c = 98.6\%$)	PASS
	10 Multi-Block Long Routes (200 m/ep, 2,000 m total)	$\geq 85.0\%$	90.0% (9/10, $R_c = 95.4\%$)	100.0% (10/10, $R_c = 100.0\%$)	PASS
	ECE: Ep-Block Val (7.5k) / Unseen Closed-Loop Test	≤ 0.0500	0.0142 (Unseen Vision Test)	0.0060 $\rightarrow$ 0.0000 (Val / GT Test)	PASS
	Aux Vision Lead Depth MAE / Cross-Track Error $\overline{\lvert e_y \rvert}$	≤ 5.0 m / ≤ 0.25 m	3.23 m / 0.094 m	3.23 m / 0.085 m	PASS
MetaDrive 0.4.3 (50 Unseen Maps)	Single-Frame Latency (Seeds 5000–5049, 10,480 m total)	≤ 40.0 ms	30.20 ms (33.1 FPS, P95: 31.67 ms)	30.20 ms (33.1 FPS)	PASS
	Destination Arrival (50 Unseen Topologies, 4,153 Frames)	$\geq 85.0\%$	96.0% (48/50 arrived; 97.4% arc)	100.0% (50/50 arrived; 98.7% arc)	PASS
	Unseen Closed-Loop Test ECE ($M = 15$ Bins)	≤ 0.0500	0.0184 (Noisy Sensor)	0.0000 (GT Teacher)	PASS

Table 4: Architecture, baseline, latency-injection, and leave-one-out controller ablation in CARLA 0.9.16 (10 scenarios).

Part A: Decision & Perception (Fixed Controller)	Arrived	R_c	$\overline{\lvert e_y \rvert}$ (m)	$\lvert \Delta\delta \rvert_{\mathrm{RMS}}$	AEB
Keep-Lane Rule (`rule_only`, `gt`)	6/10	76.4%	0.142	0.0195	58
Heuristic Overtake FSM (`heuristic_fsm`, `gt`)	8/10	88.5%	0.126	0.0284	64
Direct `ExpertLabeler` Rules (`vision_estimated`)	8/10	89.1%	0.131	0.0296	61
DINOv2 + 2-Layer MLP Head (No 421M Laya, `vision_est`)	8/10	89.4%	0.118	0.0265	56
DINOv2+Laya w/o Temporal ΔV_t (`no_temporal`, `vision_est`)	9/10	94.8%	0.102	0.0244	52
DINOv2+Laya + 100 ms Latency Injection (2-frame lag)	9/10	93.5%	0.115	0.0258	54
DINOv2+Laya + 300 ms Latency Injection (6-frame lag)	7/10	83.2%	0.168	0.0315	72
DINOv2+Laya (`vision_estimated`, Shield OFF)	9/10	94.2%	0.098	0.0241	44
DINOv2+Laya (`vision_estimated`, Shield ON)	10/10	100.0%	0.094	0.0231	49
DINOv2+Laya (`gt` State, Shield OFF, `model_only`)	10/10	100.0%	0.086	0.0226	45
DINOv2+Laya (`gt` State, Shield ON, Full)	10/10	100.0%	0.085	0.0223	46
Part B: Leave-One-Out Controller Ablation (Fixed DINOv2+Laya)	**Arrived**	R_c	$\overline{\lvert e_y \rvert}$ (m)	$\lvert \Delta\delta \rvert_{\mathrm{RMS}}$	AEB
Full Controller Stack (Quintic + Chord + CTH-ACC + FF)	10/10	100.0%	0.085	0.0223	46
− 6-WP Trajectory Curvature Feedforward ($\kappa_{\mathrm{traj}} = 0$)	10/10	100.0%	0.092	0.0235	48
− True CTH-ACC (Reverted to Fixed 16 m Gap)	10/10	100.0%	0.095	0.0248	112
− 28 m Spatial Quintic Spline (3.5 m Step WPs)	10/10	100.0%	0.148	0.0341	94
− Speed-Damped Linear Steer Ramp (Unramped PID)	10/10	100.0%	0.184	0.0618	118
All Removed (Discrete 2 m Step WPs + Fixed-Gap ACC)	10/10	100.0%	0.218	0.0642	138

between visual tokens and telemetry/navigation embeddings; (2) executing `ExpertLabeler` rules directly on noisy `vision_estimated` inputs (`expert_rule_policy`, without simulator `expert_maneuver` oracle) drops completion to 8/10 ($\overline{\lvert e_y \rvert} = 0.131$ m) due to threshold chattering; and (3) injecting a 6-frame (300 ms) action-queue delay into the same DINOv2+Laya policy drops completion to 7/10 on pedestrian AEB scenarios (`FSD-3, 9`). Part B reports both leave-one-out and cumulative controller ablations, isolating the effect of each module on $\overline{\lvert e_y \rvert}$, $\lvert \Delta\delta \rvert_{\mathrm{RMS}}$, and AEB count. Limitations include urban evaluation on a single map (`Town10HD_Opt`), supervision from rule-based `ExpertLabeler` targets rather than human trajectories, and 50 ms synchronous simulation without hardware-in-the-loop CAN-bus delay.

References

[1] Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Xi Chen, Krzysztof Choromanski, Tianli Ding, Danny Driess, Avinava Dubey, Chelsea Finn, et al. RT-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023.

[2] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9650–9660, 2021.

[3] Dian Chen, Brady Zhou, Vladlen Koltun, and Philipp Krähenbühl. Learning by cheating. In *Conference on Robot Learning (CoRL)*, pages 66–75, 2020.

[4] Kashyap Chitta, Aditya Prakash, Bernhard Jaeger, Zehao Yu, Katrin Renz, and Andreas Geiger. TransFuser: Imitation with transformer-based sensor fusion for autonomous driving. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 45(11):12878–12895, 2022.

[5] Felipe Codevilla, Matthias Müller, Antonio López, Vladlen Koltun, and Alexey Dosovitskiy. End-to-end driving via conditional imitation learning. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 4693–4700, 2018.

[6] ConvAI Innovations. Laya: Non-autoregressive low-latency decision models on bidirectional modernbert. https://huggingface.co/convaiinnovations/laya, 2024.

[7] R. Craig Coulter. Implementation of the pure pursuit path tracking algorithm. Technical Report CMU-RI-TR-92-01, Carnegie Mellon University, The Robotics Institute, 1992.

[8] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 4171–4186, 2019.

[9] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

[10] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, and Vladlen Koltun. CARLA: An open urban driving simulator. In *Proceedings of the 1st Annual Conference on Robot Learning (CoRL)*, pages 1–16, 2017.

[11] Marius Dupuis, Martin Strobl, and Hans Grezlikowski. OpenDRIVE format specification, rev. 1.4. *VIRES Simulationstechnologie GmbH*, 2010.

[12] Tamar Flash and Neville Hogan. The coordination of arm movements: An experimentally confirmed mathematical model. *Journal of Neuroscience*, 5(7):1688–1703, 1985.

[13] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, pages 1321–1330, 2017.

[14] Peter E. Hart, Nils J. Nilsson, and Bertram Raphael. A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2):100–107, 1968.

[15] Gabriel M. Hoffmann, Claire J. Tomlin, Michael Montemerlo, and Sebastian Thrun. Autonomous automobile trajectory tracking for off-road driving: Controller design, experimental validation and racing. *Proceedings of the American Control Conference (ACC)*, pages 2296–2301, 2007.

[16] Yihan Hu, Jiazhi Yang, Li Chen, Keyu Li, Chonghao Sima, Xizhou Zhu, Siqi Chai, Senyao Du, Tianwei Lin, Wenhai Wang, et al. Planning-oriented autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17853–17862, 2023.

[17] Petros A. Ioannou and C. C. Chien. Autonomous intelligent cruise control. *IEEE Transactions on Vehicular Technology*, 42(4):657–672, 1993.

[18] Bernhard Jaeger, Kashyap Chitta, and Andreas Geiger. Hidden biases of end-to-end driving models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8240–8249, 2023.

[19] Bo Jiang, Shaoyu Chen, Qing Xu, Bencheng Liao, Jiajie Chen, Helong Zhou, Qian Zhang, Wenyu Liu, Chang Huang, and Xinggang Wang. VAD: Vectorized scene representation for efficient autonomous driving. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 8340–8350, 2023.

[20] Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, Grace Lam, Pannag Sanketi, et al. OpenVLA: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.

[21] Quanyi Li, Zhenghao Peng, Lan Feng, Qihang Zhang, Zhenghai Xue, and Bolei Zhou. MetaDrive: Composing diverse driving scenarios for generalizable reinforcement learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 45(3):3461–3475, 2022.

[22] Zhiqi Li, Wenhai Wang, Hongyang Li, Enze Xie, Chonghao Sima, Tong Lu, Yu Qiao, and Jifeng Dai. BEVFormer: Learning bird's-eye-view representation from multi-camera images via spatiotemporal transformers. In *European Conference on Computer Vision (ECCV)*, pages 1–18, 2022.

[23] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

[24] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[25] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research (TMLR)*, 2024.

[26] Brian Paden, Michal Čáp, Sze Zheng Yong, Dmitry Yershov, and Emilio Frazzoli. A survey of motion planning and control techniques for self-driving urban vehicles. *IEEE Transactions on Intelligent Vehicles*, 1(1):33–55, 2016.

[27] Aditya Prakash, Kashyap Chitta, and Andreas Geiger. Multi-modal fusion transformer for end-to-end autonomous driving. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7077–7087, 2021.

[28] Rajesh Rajamani and Chuan Zhu. An experimental comparative study of autonomousogh and cooperative vehicle-following control laws. *IEEE Transactions on Control Systems Technology*, 10(3):371–383, 2002.

[29] Sebastian Thrun, Michael Montemerlo, Hendrik Dahlkamp, David Stavens, Andrei Aron, James Diebel, Philip Fong, John Gale, Morgan Halpenny, Gabriel Hoffmann, et al. Stanley: The robot that won the DARPA grand challenge. *Journal of Field Robotics*, 23(9):661–692, 2006.

[30] Martin Treiber, Ansgar Hennecke, and Dirk Helbing. Congested traffic states in empirical observations and microscopic simulations. *Physical Review E*, 62(2):1805–1824, 2000.

[31] Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom

Aarsen, et al. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. *arXiv preprint arXiv:2412.13663*, 2024.

[32] Moritz Werling, Julius Ziegler, Sören Kammel, and Sebastian Thrun. Optimal trajectory generation for dynamic street scenarios in a Frenet frame. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 987–993, 2010.